



Written examination: 28th May 2025, 9:00-13:00.

Course number: 02450

Course name: Introduction to Machine Learning and Data Mining

Aids allowed: No aids, except two A4 sheets with handwritten notes (front and back), standard dictionaries, and a non-programmable calculator.

Exam duration: 4 hours.

Format: The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. A correct answer gives 3 points, a wrong answer gives -1 point, and “Don’t know” (E) gives 0 points.

Weighting: Exam questions are weighted equally.

Answers This exam only allows for physical hand-in. Select **one** answer per exam question by completely filling in a circle (e.g.     ) using a black/dark pencil or pen. Do **not** use checkmarks or crosses to indicate your desired answer. To change an answer either erase the previously marked answer using an eraser, or clearly cross over the answer and fill another circle and write the letter of the final answer at the end of the line, e.g.                                   

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	Attribute description	Abbrev.
x_1	Main motor temperature (in degrees celcius)	MTEMP
x_2	Force	FORCE
x_3	Vibration	VIB
x_4	Angular speed (in degrees per second)	SPEED
x_5	Ambient temperature (in degrees celcius)	ATEMP
y	Operational status	STATUS
y_r	Power consumption	PC

Table 1: Description of the attributes of the Welding robot dataset used in this exam. The dataset used in this exam relates to a large mechanical welding robot used to construct container ships at a shipyard. Measurements have been made to model the robot's power consumption, $y_r \in [0, \infty[$, and the operational status, $y \in \{1, 2\}$, where $y = 1$ indicates an Ideal mode of operation and $y = 2$ indicates a Normal mode of operation. The dataset contains $N = 500$ observations.

Question 1. The main dataset used in this exam is the Welding robot dataset described in Table 1. We consider the first four attributes of the Welding robot dataset in Table 1. A scatter plot matrix for these attributes is given in Figure 1.

Which one of the following matrices represents the empirical correlation matrix for these attributes?

A. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.01 & -0.15 & 0.01 \\ 0.01 & 1.0 & 0.18 & 0.27 \\ -0.15 & 0.18 & 1.0 & 0.82 \\ 0.01 & 0.27 & 0.82 & 1.0 \end{bmatrix}$

B. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.01 & -0.15 & 0.01 \\ 0.01 & 1.0 & 0.18 & 0.27 \\ -0.15 & 0.18 & 1.0 & 0.16 \\ 0.01 & 0.27 & 0.16 & 1.0 \end{bmatrix}$

C. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.01 & -0.85 & 0.01 \\ 0.01 & 1.0 & 0.18 & 0.27 \\ -0.85 & 0.18 & 1.0 & 0.82 \\ 0.01 & 0.27 & 0.82 & 1.0 \end{bmatrix}$

D. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.01 & -0.15 & 0.01 \\ 0.01 & 1.0 & 0.18 & 0.27 \\ -0.15 & 0.18 & 1.0 & -0.82 \\ 0.01 & 0.27 & -0.82 & 1.0 \end{bmatrix}$

E. Don't know.

Question 2. We consider the histograms of the four attributes in Figure 2. Match the histograms to the attributes by using the information regarding the distributions of the points from Figure 1. Which one

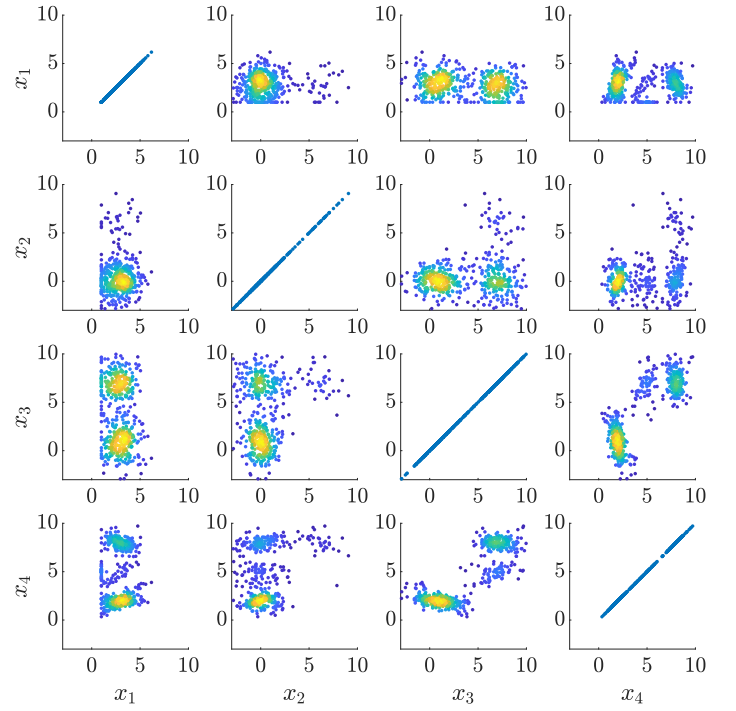


Figure 1: The scatter plot matrix for the attributes x_1 , x_2 , x_3 and x_4 of the Welding robot dataset of Table 1. In the off-diagonal panels, the color represents the associated density.

of the following statements is *correct*?

- A. Histogram 1: x_1 , Histogram 2: x_2
Histogram 3: x_4 , Histogram 4: x_3 .
- B. Histogram 1: x_4 , Histogram 2: x_2
Histogram 3: x_3 , Histogram 4: x_1 .
- C. Histogram 1: x_3 , Histogram 2: x_1
Histogram 3: x_2 , Histogram 4: x_4 .
- D. Histogram 1: x_4 , Histogram 2: x_1
Histogram 3: x_2 , Histogram 4: x_3 .
- E. Don't know.

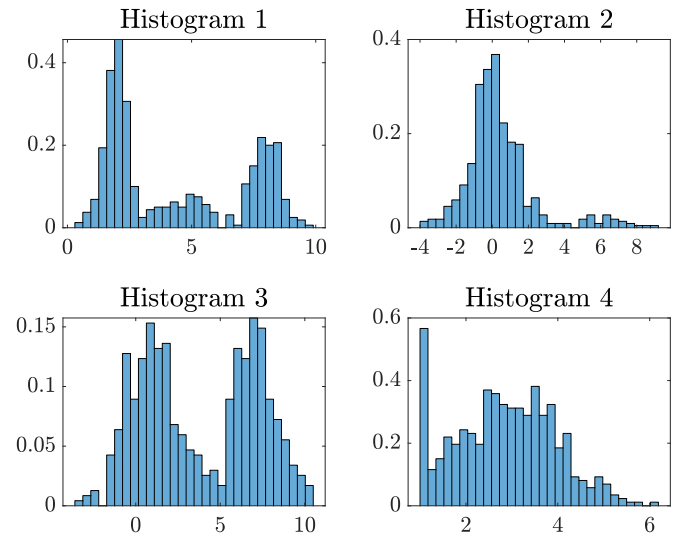


Figure 2: Histograms of the attributes in which their order has been permuted.

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
o_1	0.0	10.0	11.8	8.1	9.9	6.6	10.7	2.3	7.5	7.4
o_2	10.0	0.0	13.7	4.3	9.5	7.0	8.7	8.4	5.4	6.2
o_3	11.8	13.7	0.0	11.3	5.8	10.2	7.0	10.9	9.0	8.4
o_4	8.1	4.3	11.3	0.0	8.7	3.3	7.5	5.9	2.5	3.0
o_5	9.9	9.5	5.8	8.7	0.0	8.4	3.4	9.0	6.5	6.3
o_6	6.6	7.0	10.2	3.3	8.4	0.0	7.9	4.5	3.0	2.9
o_7	10.7	8.7	7.0	7.5	3.4	7.9	0.0	9.3	5.9	5.7
o_8	2.3	8.4	10.9	5.9	9.0	4.5	9.3	0.0	5.5	5.4
o_9	7.5	5.4	9.0	2.5	6.5	3.0	5.9	5.5	0.0	0.8
o_{10}	7.4	6.2	8.4	3.0	6.3	2.9	5.7	5.4	0.8	0.0

Table 2: The pairwise Euclidian distances, $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$ between 10 observations from the Welding robot dataset (recall that $M = 5$). Each observation o_i corresponds to a row of the data matrix $\mathbf{X}$ corresponding to the attributes in Table 1. The colors indicate classes such that the black observations $\{o_1, o_2, o_3\}$ belong to class C_1 (corresponding to Ideal), and the red observations $\{o_4, o_5, o_6, o_7, o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to Normal).

Question 3. Consider the distances in Table 2 based on 10 observations from the Welding robot dataset. The class labels C_1, C_2 (see table caption for details) will be predicted using a k -nearest neighbour classifier based on the distances given in Table 2. Suppose we use leave-one-out cross validation (i.e., the observation that is being predicted is left out) and a 1-nearest neighbour classifier (i.e., $k = 1$). What is the error rate computed for all $N = 10$ observations?

- A. error rate = 1
- B. error rate = $\frac{9}{10}$
- C. error rate = $\frac{2}{5}$
- D. error rate = $\frac{1}{5}$
- E. Don't know.

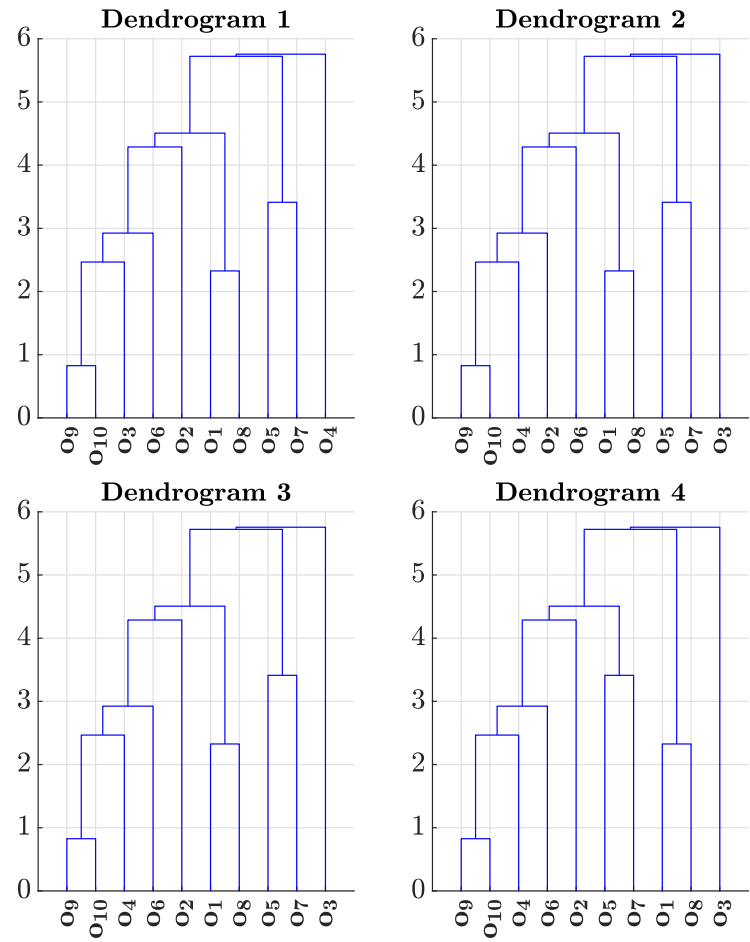


Figure 3: Proposed hierarchical clustering of the 10 observations in Table 2.

Question 4. A hierarchical clustering is applied to the 10 observations in Table 2 using *minimum* linkage. Which one of the dendrograms shown in Figure 3 corresponds to the distances given in Table 2?

- A. Dendrogram 1
- B. Dendrogram 2
- C. Dendrogram 3
- D. Dendrogram 4
- E. Don't know.

Question 5. Consider the Principal Component Analysis (PCA) of the Welding robot dataset represented by the $N \times M$ data matrix $\mathbf{X}$ with the observations as row vectors, i.e., the i 'th row in $\mathbf{X}$ is $\mathbf{x}_i^\top$.

The Singular Value Decomposition (SVD) is given by $\tilde{\mathbf{X}} = \mathbf{U}\Sigma\mathbf{V}^\top$, where $\tilde{\mathbf{X}}$ is the centered data matrix obtained from $\mathbf{X}$ by subtracting the column-wise mean. The projection of $\mathbf{X}$ onto a principal component results in N scalar values denoted by $b = \{b_1, b_2, \dots, b_N\}$. Determine which one of the following options results in an unbiased estimate of the variance of b when projecting $\mathbf{X}$ onto the second principal component.

- A. $\text{Var}[b]$ is *always* 1
- B. $\text{Var}[b] = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i^\top \mathbf{u}_2)^2$
- C. $\text{Var}[b] = \frac{1}{N-1} \mathbf{v}_2^\top \tilde{\mathbf{X}}^\top \mathbf{v}_2$
- D. $\text{Var}[b] = \frac{1}{N-1} \sum_{i=1}^N \left[\left(\mathbf{x}_i^\top - \frac{1}{N} \sum_{j=1}^N \mathbf{x}_j^\top \right) \mathbf{v}_2 \right]^2$
- E. Don't know.

Question 6. Consider again the Principal Component Analysis of the Welding robot dataset, and in particular the SVD decomposition of the centered data matrix $\tilde{\mathbf{X}} = \mathbf{U}\Sigma\mathbf{V}^\top$, and the corresponding matrix

$$\mathbf{V} = \begin{bmatrix} 0.01 & 0.03 & -0.95 & -0.22 & 0.23 \\ 0.31 & -0.67 & -0.07 & 0.57 & 0.35 \\ 0.64 & 0.05 & 0.25 & -0.55 & 0.47 \\ 0.68 & 0.06 & -0.19 & 0.11 & -0.7 \\ 0.19 & 0.74 & -0.02 & 0.56 & 0.33 \end{bmatrix}.$$

Which one of the following statements is *correct*?

- A. An observation with a low value of x_2 (**FORCE**), a low value of x_3 (**VIB**), and a low value of x_4 (**SPEED**) will typically have a positive value of the projection onto principal component number 1.
- B. An observation with a high value of x_2 (**FORCE**), and a low value of x_5 (**ATEMP**) will typically have a negative value of the projection onto principal component number 2.
- C. An observation with a high value of x_1 (**MTEMP**), a low value of x_2 (**FORCE**), a high value of x_3 (**VIB**), and a low value of x_5 (**ATEMP**) will typically have a positive value of the projection onto principal component number 4.
- D. An observation with a high value of x_1 (**MTEMP**), and a low value of x_3 (**VIB**) will typically have a positive value of the projection onto principal component number 3.
- E. Don't know.

	f_1	f_2	f_3	f_4	f_5
o_1	0	1	1	0	1
o_2	0	1	0	0	1
o_3	1	1	1	0	1
o_4	0	0	0	0	1
o_5	0	1	1	1	1
o_6	0	0	1	1	1
o_7	0	0	0	1	1
o_8	0	0	1	1	1
o_9	0	1	0	0	1
o_{10}	0	0	1	1	1

Table 3: Binarized version of the Welding robot dataset. Each of the features f_i are obtained by taking a feature x_i and letting $f_i = 1$ correspond to a value x_i greater than the median (otherwise $f_i = 0$). The colors indicate classes such that the black observations $\{o_1, o_2, o_3\}$ belong to class C_1 (corresponding to Ideal), and the red observations $\{o_4, o_5, o_6, o_7, o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to Normal).

Question 7. We consider the binary matrix from Table 3 as a market basket problem consisting of $N = 10$ transactions $o_1, \dots, o_{10}$ and $M = 5$ items $f_1, \dots, f_5$.

Determine how many itemsets that have *support* $\epsilon > 0.55$.

- A. 32
- B. 9
- C. 6
- D. 3
- E. Don't know.

Question 8. Consider a Naïve Bayes classifier for determining if $y = 1$ (Ideal operation) or $y = 2$ (Normal operation) based on the binarized version of the Welding robot dataset shown in Table 3.

Determine which one of the following statements is correct.

- A. $p(y = 2 \mid f_2 = 1) = p(y = 2 \mid f_2 = 1, f_5 = 1)$
- B. $p(y = 1 \mid f_1 = 0, f_2 = 1, f_3 = 1, f_4 = 0) = 0.0$
- C. $p(y = 2 \mid f_2 = 1) = \frac{1}{5}$
- D. $p(y = 2 \mid f_2 = 1) = p(y = 2 \mid f_2 = 0)$
- E. Don't know.

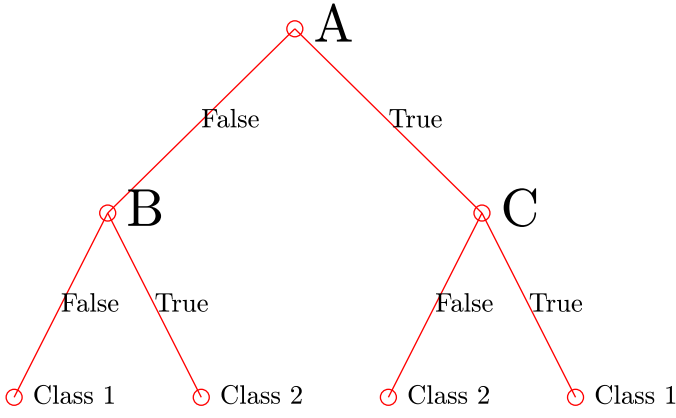


Figure 4: Classification tree.

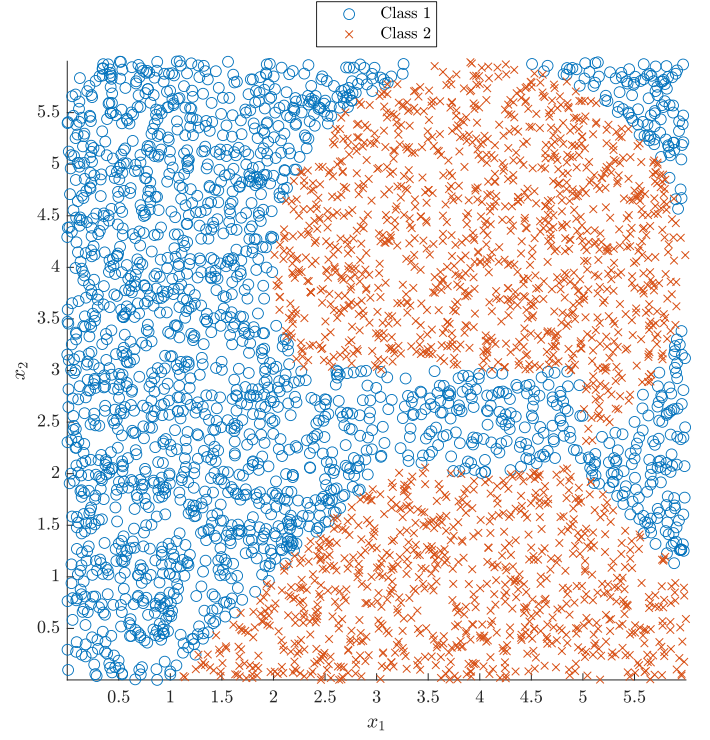


Figure 5: Classification boundary.

Question 9. Consider a small dataset comprised of $N = 7$ observations

$$x = [3 \ 4 \ 6 \ 13 \ 16 \ 18 \ 20].$$

Suppose a k -means algorithm is applied to the dataset with $K = 3$ clusters and using Euclidian distances. At a given stage of the algorithm the data is partitioned into the blocks:

$$\{3, 4\}, \{6, 13, 16\}, \{18, 20\}$$

What clustering will the k -means algorithm eventually converge to?

- A. $\{\}, \{3, 4, 6, 13, 16\}, \{18, 20\}$
- B. $\{3, 4, 6\}, \{13\}, \{16, 18, 20\}$
- C. $\{3, 4\}, \{6, 13, 16, 18\}, \{20\}$
- D. $\{3, 4\}, \{6, 13\}, \{16, 18, 20\}$
- E. Don't know.

Question 10. We consider an artificial dataset of $N = 4000$ observations. The dataset is classified according to a decision tree of the form shown in Figure 4 resulting in a partition into classes indicated by the colors/markers in Figure 5. What is the correct

rule assignment to the nodes in the decision tree?

- A. $\mathbf{A}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right\|_1 < 3, \mathbf{B}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \end{bmatrix} \right\|_2 < 2,$
 $\mathbf{C}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_\infty < 3$
- B. $\mathbf{A}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \end{bmatrix} \right\|_2 < 2, \mathbf{B}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right\|_1 < 3,$
 $\mathbf{C}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_\infty < 3$
- C. $\mathbf{A}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_\infty < 3, \mathbf{B}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \end{bmatrix} \right\|_2 < 2,$
 $\mathbf{C}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right\|_1 < 3$
- D. $\mathbf{A}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right\|_1 < 3, \mathbf{B}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\|_\infty < 3,$
 $\mathbf{C}: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \end{bmatrix} \right\|_2 < 2$
- E. Don't know.

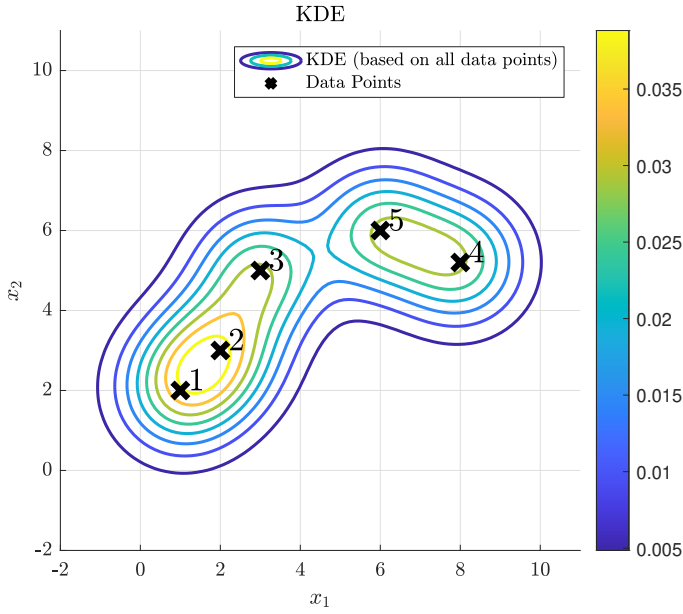


Figure 6: KDE

Obs.	Component				
	1	2	3	4	5
$\mathbf{x}_1$	0.13	0.06	0.00	0.00	0.00
$\mathbf{x}_2$	0.06	0.13	0.02	0.00	0.00
$\mathbf{x}_3$	0.00	0.02	0.13	0.00	0.00
$\mathbf{x}_4$	0.00	0.00	0.00	0.13	0.02
$\mathbf{x}_5$	0.00	0.00	0.00	0.02	0.13

Table 4: Evaluation of each *individual* component/kernel at each observation.

Question 11. Figure 6 shows a kernel density estimator (KDE) on five datapoints using the standard formulation of the KDE, i.e.,

$$p(\mathbf{x}) = \frac{1}{N} \sum_{n=1}^N \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_n, \boldsymbol{\Sigma} = 2\mathbf{I})$$

Table 4 shows the evaluation of each *individual* component/kernel at each of the five observations. Compute the leave-one-out (LOO) estimate of $p(\mathbf{x}_2)$.

- A. $p(\mathbf{x}_2) \approx 0.016$
- B. $p(\mathbf{x}_2) \approx 0.020$
- C. $p(\mathbf{x}_2) \approx 0.053$
- D. $p(\mathbf{x}_2) \approx 0.080$
- E. Don't know.

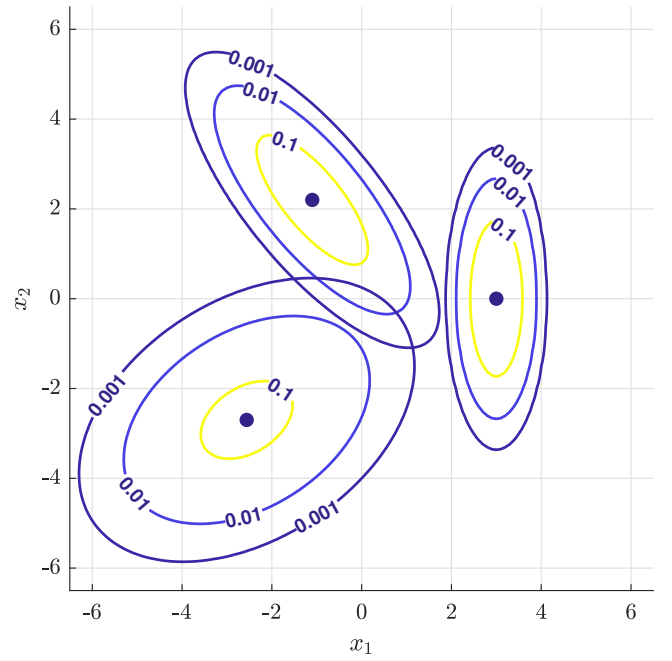


Figure 7: Contour lines of the densities corresponding to the three individual components.

Question 12. A Gaussian Mixture Model with $K = 3$ components has been trained on a dataset with two attributes. The trained model is defined as

$$p(\mathbf{x}) = \sum_{k=1}^3 p(z = k) \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

with the following covariance matrices:

$$\boldsymbol{\Sigma}_1 = \begin{bmatrix} 0.7 & -0.62 \\ -0.62 & 0.95 \end{bmatrix}, \quad \boldsymbol{\Sigma}_2 = \begin{bmatrix} 1.4 & 0.45 \\ 0.45 & 1.0 \end{bmatrix},$$

$$\boldsymbol{\Sigma}_3 = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.9 \end{bmatrix}.$$

The prior probability of the components are equal, i.e., $p(z = 1) = p(z = 2) = p(z = 3) = \frac{1}{3}$. The contour lines of the densities corresponding to the individual mixture components are shown in Figure 7. Given a test point $\mathbf{x} = [0 \ 0]^\top$, determine which component the test point is most likely to belong to, i.e., the component that maximizes $p(z = k \mid \mathbf{x})$.

- A. Component 1
- B. Component 2
- C. Component 3
- D. The point has equal probability of belonging to the three components
- E. Don't know.

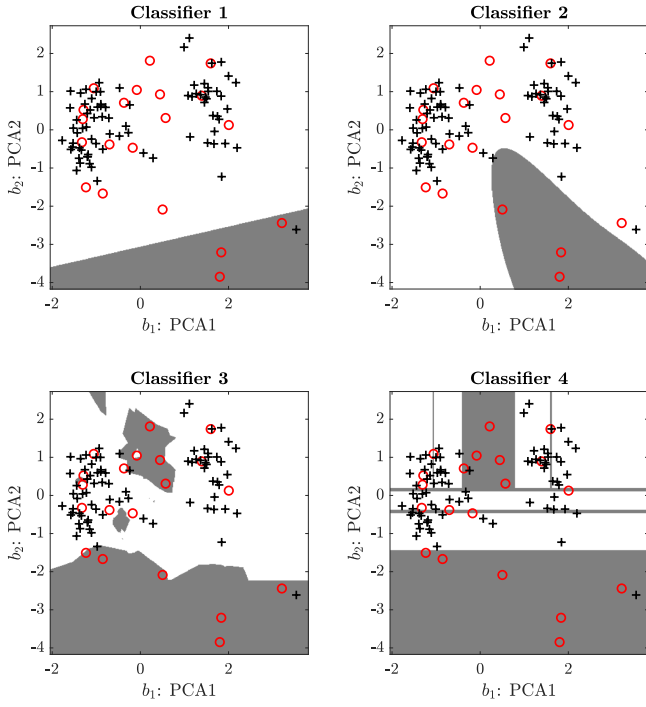


Figure 8: Decision boundaries for four different classifiers trained on the Welding robot dataset when projected onto the first two principal components (only a subset of the observations is shown).

Question 13. We will consider a subset of the Welding robot dataset (described in Table 1) after it has been projected onto the first two principal components b_1 and b_2 , thereby giving rise to a smaller two-dimensional dataset. We will consider the following four classifiers:

LREG: Logistic regression

ANN: Artificial neural network with 5 hidden units

DT: Decision tree with regular axis-aligned splits ($b_i < c$)

KNN: K-nearest neighbours with $K = 3$

The classifiers are trained on a two-dimensional dataset and the decision boundary for each of the four classifiers is given in Figure 8. Which one of the following statements is correct?

- A. Classifier 1 corresponds to **LREG**,
Classifier 2 corresponds to **KNN**,
Classifier 3 corresponds to **ANN**,
Classifier 4 corresponds to **DT**.
- B. Classifier 1 corresponds to **DT**,
Classifier 2 corresponds to **LREG**,
Classifier 3 corresponds to **KNN**,
Classifier 4 corresponds to **ANN**.
- C. Classifier 1 corresponds to **LREG**,

Classifier 2 corresponds to **ANN**,
Classifier 3 corresponds to **KNN**,
Classifier 4 corresponds to **DT**.

- D. Classifier 1 corresponds to **ANN**,
Classifier 2 corresponds to **DT**,
Classifier 3 corresponds to **KNN**,
Classifier 4 corresponds to **LREG**.

E. Don't know.

Question 14. Consider a regularized linear regression model trained with the regularization parameter/strength $\lambda = 1.49$ on the Welding robot dataset using the five attributes in addition to third-order polynomial terms as input, i.e.

$$\tilde{\mathbf{x}} = [1, x_1, x_2, \dots, x_5, x_1^3, x_2^3, \dots, x_5^3]^\top.$$

The model performs poorly on a held-out test set and has been confirmed to have low bias but very high variance.

Which one of the following actions is most likely to improve the generalization performance on unseen data?

- A. Add forth-order polynomial transformations of the data.
- B. Decrease the amount of regularization applied to the model.
- C. Train only on half the training set with $\lambda = 1.49$.
- D. Increase the value of λ .
- E. Don't know.

Question 15. Which one of the following statements about cross-validation (CV) is *correct*?

- A. In hold-out cross-validation, using the same test set for learning the complexity of the model and for the final evaluation of the performance gives an accurate estimate of the generalization error.
- B. In K -fold cross-validation the trained models are more similar to each other compared to those in leave-one-out cross-validation because the overlap between the training sets is larger.
- C. For a given dataset, leave-one-out cross-validation typically gives a better estimate of the generalization error than K -fold cross-validation and hold-out cross-validation.
- D. The two-layer cross-validation procedure can be used only for supervised and not unsupervised learning problems.
- E. Don't know.

Question 16. Consider clustering $N = 4$ observations with the ground-truth given by each observation belonging to its own cluster such that the partition Q becomes:

$$Q = [1 \quad 2 \quad 3 \quad 4]$$

The elements in the vector indicate the cluster index for a specific observation.

We use the *Rand index* to compare two partitions, Q and Z , i.e.,

$$Rand(Q, Z) = \frac{S + D}{\frac{1}{2} \cdot N \cdot (N - 1)},$$

where S is the number of times Z and Q agree that two observations are in the same cluster in both partitions. D is the number of times Z and Q agree that two observations are not in the same cluster in both partitions.

Determine which one of the following options for the Z partition that results in the the highest *Rand index* when compared to Q .

- A. $Z^{(A)} = [1 \quad 1 \quad 2 \quad 2]$
- B. $Z^{(B)} = [1 \quad 2 \quad 3 \quad 2]$
- C. $Z^{(C)} = [1 \quad 2 \quad 1 \quad 2]$
- D. $Z^{(D)} = [1 \quad 2 \quad 2 \quad 2]$
- E. Don't know.

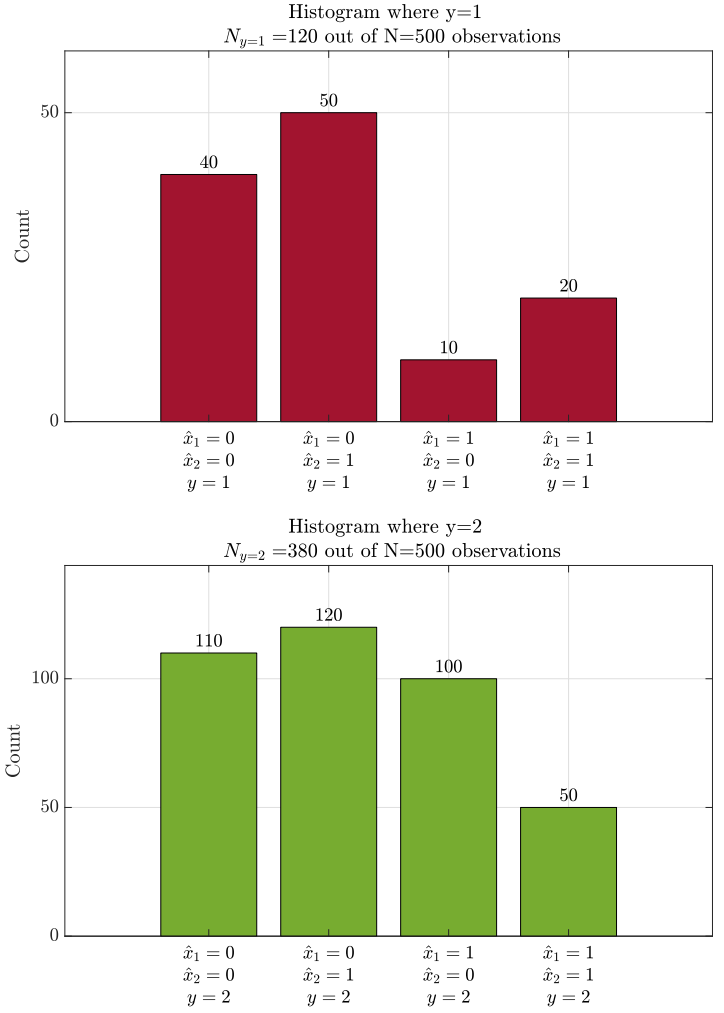


Figure 9: Histogram of a binarized version of the Welding robot dataset described in Table 1.

Question 17. Consider the Welding robot dataset from Table 1. The attributes have been binarized to produce new features denoted $\hat{x}_1, \hat{x}_2, \dots, \hat{x}_M$. A summary of the binarised dataset is given in Figure 9 for a subset of the (binarized) attributes and label, namely $\hat{x}_1, \hat{x}_2$ and y .

Determine the probability of $y = 2$ given an observation has $\hat{x}_1 = 1$ and $\hat{x}_2 = 0$ using Bayes theorem (without applying the Naïve Bayes assumption).

- A. 0.691
- B. 0.909
- C. 0.263
- D. 0.379
- E. Don't know.

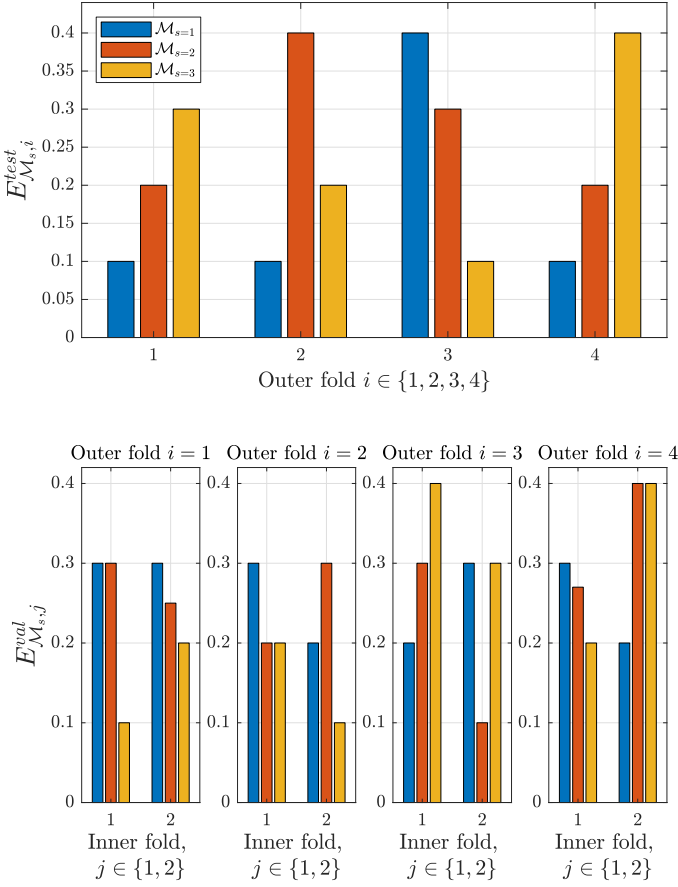


Figure 10: Result of running two-layer cross-validation.

Question 18. Consider evaluating three regression models ($\mathcal{M}_{s=1}$, $\mathcal{M}_{s=2}$, and $\mathcal{M}_{s=3}$) of different complexity, based on a dataset with 1200 observations. We use two-layer cross-validation with $K_1 = 4$ outer folds and $K_2 = 2$ inner folds. The error is computed using the mean squared error.

The result of the procedure is shown in Figure 10. $E_{\mathcal{M}_{s,i}}^{test}$ denotes the performance of model s on the test set in the i^{th} outer fold. $E_{\mathcal{M}_{s,j}}^{val}$ denotes the performance of model s on the validation set in the j^{th} inner fold. Hence, we do not just choose the optimal model, $\mathcal{M}_{s^*}$, based on the result of the inner fold, but compute the error for all three models on the test set in each of the outer folds.

Determine the value of the estimated generalization error $\hat{E}^{gen}$.

- A. $\hat{E}^{gen} \approx 0.3$
- B. $\hat{E}^{gen} \approx 0.1$
- C. $\hat{E}^{gen} \approx 0.25$
- D. $\hat{E}^{gen} \approx 0.225$
- E. Don't know.

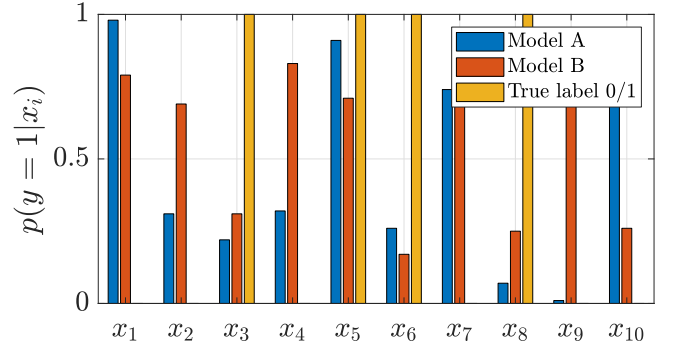


Figure 11: $p(y = 1|x_i)$ for the two models on a test set of 10 points $x_1, \dots, x_{10}$, and the true labels. If $p(y = 1|x_i) > 0.5$, then x_i is classified as Class 1 and otherwise Class 0.

Question 19. Consider a generic binary classification problem on which two classifiers, Model A and Model B, have been trained to predict if an observation, x_i , belongs to Class 0 or Class 1. We wish to compare the two models using McNemar's test to determine if there is a significant difference between them. The probabilities $p(y = 1 | x_i)$ for the two models on a test set of $n = 10$ points are shown in Figure 11, together with the true test label. Our null hypothesis is that the classifiers perform similarly.

Determine which combination of estimated difference in accuracy $\hat{\theta}$, confidence interval (CI) with $\alpha = 0.05$, and p-value is the only plausible answer.

Hint: Generic information about McNemar's test for comparing the accuracy difference $\theta = \theta_A - \theta_B$ of two classifiers.

$$p(\theta | \mathbf{n}) = \frac{1}{2} \text{Beta} \left(\frac{\theta + 1}{2} \mid a = f, b = g \right)$$

where $f = \frac{E_\theta + 1}{2}(Q - 1)$, $g = \frac{1 - E_\theta}{2}(Q - 1)$ and $E_\theta = \frac{n_{12} - n_{21}}{n}$.

$$Q = \frac{n^2(n + 1)(E_\theta + 1)(1 - E_\theta)}{n(n_{12} + n_{21}) - (n_{12} - n_{21})^2}$$

$$\theta_L = 2 \text{cdf}_{\text{Beta}}^{-1} \left(\frac{\alpha}{2} \mid a = f, b = g \right) - 1$$

$$\theta_U = 2 \text{cdf}_{\text{Beta}}^{-1} \left(1 - \frac{\alpha}{2} \mid a = f, b = g \right) - 1$$

$\mathbf{n}$ is a 2×2 matrix with n_{11} and n_{22} being the number of times model A and model B are simultaneously correct and wrong, respectively. n_{12} is the number of times model A is correct but model B wrong and vice-versa for n_{21} .

- A. $\hat{\theta} = 0.2$, $CI = [-0.16, 0.54]$, p-value > 0.5
- B. $\hat{\theta} = -0.36$, $CI = [-0.16, 0.54]$, p-value < 0.63
- C. $\hat{\theta} = 0.2$, $CI = [-0.16, 0.54]$, p-value < 0.01
- D. $\hat{\theta} = -0.4$, $CI = [-1.0, 0.33]$, p-value < 0.63
- E. Don't know.

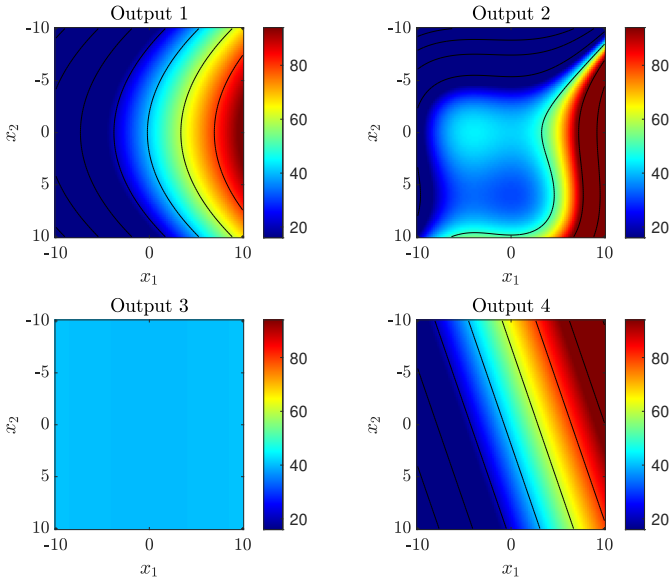


Figure 12: Functions are illustrated as heatmaps with contour lines overlayed.

Question 20. Figure 12 shows the output of four different regularized linear regression models. We use the mean squared error to measure the discrepancy between predictions and target values. The weights are penalized using the Euclidian norm and regularization parameter/strength λ . The weight corresponding to the intercept/offset term is not regularized. The models are trained to predict the output based on transformations of x_1 and x_2 and using the standard prediction rule $f(\mathbf{x}) = \mathbf{w}^\top \tilde{\mathbf{x}}$. The models are given by:

- Model A: $\tilde{\mathbf{x}} = [1 \ x_1 \ x_2]^\top$ and $\lambda = 0.5$.
 Model B: $\tilde{\mathbf{x}} = [1 \ x_1^2 \ x_2^2]^\top$ and $\lambda = 1 \cdot 10^6$.
 Model C: $\tilde{\mathbf{x}} = [1 \ x_1 \ x_2^2]^\top$ and $\lambda = 0.0$.
 Model D: $\tilde{\mathbf{x}} = [1 \ x_1^2 \ x_2^2 \ x_1^3 \ x_2^3]^\top$ and $\lambda = 0.5$.

Determine which of the following options is correct.

- A. Model A corresponds to Output 1
 Model B corresponds to Output 2
 Model C corresponds to Output 3
 Model D corresponds to Output 4
- B. Model A corresponds to Output 2
 Model B corresponds to Output 3
 Model C corresponds to Output 4
 Model D corresponds to Output 1
- C. Model A corresponds to Output 2
 Model B corresponds to Output 1
 Model C corresponds to Output 3
 Model D corresponds to Output 4
- D. Model A corresponds to Output 4
 Model B corresponds to Output 3
 Model C corresponds to Output 1
 Model D corresponds to Output 2
- E. Don't know.

Obs.	$y^{(actual)}$	$\hat{y}_{t=1}$	$\hat{y}_{t=2}$	$\hat{y}_{t=3}$
$\mathbf{x}_1$	1	2	2	1
$\mathbf{x}_2$	2	2	1	2
$\mathbf{x}_3$	1	1	1	1
$\mathbf{x}_4$	1	2	1	1
$\mathbf{x}_5$	2	1	2	2
$\mathbf{x}_6$	2	2	2	1

Table 5: For each of the $N = 6$ observations (first column), the table indicate the true class labels y^{actual} (second column) and the predicted outputs of the AdaBoost classifier $\hat{y}_t$ for the first three, $t = \{1, 2, 3\}$, rounds of boosting.

Question 21. Consider again the Welding robot dataset of Table 1. We limit ourselves to $N = 6$ arbitrary observations from the original dataset and only the features x_1 and x_3 to predict the class $y = 1$ or $y = 2$. We use a KNN classification model ($K = 1$) and apply the AdaBoost algorithm. We run AdaBoost for three rounds, initially with equal weights for each data point, i.e., $\mathbf{w}(t = 1) = [\frac{1}{6} \ \frac{1}{6} \ \frac{1}{6} \ \frac{1}{6} \ \frac{1}{6} \ \frac{1}{6}]$, resulting in the predictions tabulated in Table 5 alongside the actual/true labels.

Given this information, determine the (unnormalized) weights after round $t = 2$ of AdaBoost.

Hint: Generic information related to AdaBoost, the exponential function, and the natural logarithm. In AdaBoost, the weighted error rate is

$$\varepsilon_t = \sum_{i=1}^N w_i(t) (1 - \delta_{f_t(\mathbf{x}_i), y_i}) .$$

The update rule for the unnormalized weights is:

$$\tilde{w}_i(t+1) = \begin{cases} w_i(t)e^{-\alpha_t}, & \text{if } f_t(\mathbf{x}_i) = y_i \\ w_i(t)e^{\alpha_t}, & \text{if } f_t(\mathbf{x}_i) \neq y_i \end{cases}$$

where

$$\alpha_t = \frac{1}{2} \log \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right) .$$

AdaBoost predictions at timestep T are made based on $\hat{y}_T = f^*(\mathbf{x}) = \arg \max_{y=1,2} \sum_{t=1}^T \alpha_t \delta_{f_t(\mathbf{x}), y}$. Here $\log(x)$ consistently refers to the natural logarithm (i.e., base e). Note that $e^{c \log(z)} = z^c$.

- A. $\mathbf{w}(t = 3) \propto \frac{1}{6} \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
- B. $\mathbf{w}(t = 3) \propto \frac{1}{6} \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{bmatrix}$
- C. $\mathbf{w}(t = 3) \propto \frac{1}{6} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 2 & 2 & 2 & 2 \end{bmatrix}$
- D. $\mathbf{w}(t = 3) \propto \frac{1}{6} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \sqrt{2} & \sqrt{2} & \sqrt{2} & \sqrt{2} \end{bmatrix}$
- E. Don't know.

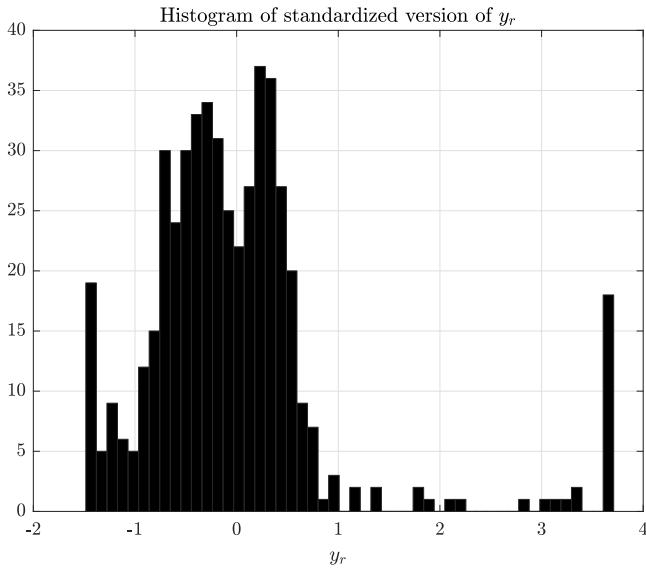


Figure 13: Power consumption y_r (standardized).

Question 22. Consider an artificial neural network (ANN) for predicting the standardized variable, y_r , measuring the power consumption in the Welding robot dataset. The ANN uses attributes $x_1, x_2, \dots, x_5$, and has one hidden layer with 256 hidden units and one output neuron. The parameters in the ANN are optimized based on the mean squared error loss function. Figure 13 shows the distribution of the standardized y_r attribute to be predicted.

Determine which one of the following activation functions would be the most suitable to use for the output neuron/unit in the ANN.

- A. A constant function: $h(z) = c$ where $c \in \mathbb{R}$ is a constant.
- B. Linear function: $h(z) = z$.
- C. Rectified linear unit (ReLU): $h(z) = \max(0, z)$.
- D. Logistic sigmoid: $h(z) = \frac{1}{1+e^{-z}}$.
- E. Don't know.

Question 23. We will consider an artificial neural network (ANN) trained on the Welding robot dataset described in Table 1 to predict the class label y based on attributes $x_1, \dots, x_5$. The neural network has a single hidden layer containing $n_h = 4$ units, and uses the sigmoid activation function to predict the class label y . For the hidden layer we will use the tanh non-linear activation function. How many parameters including those related to bias/offset terms has to be trained to fit the neural network?

- A. Network contains 29 parameters
- B. Network contains 28 parameters
- C. Network contains 32 parameters
- D. Network contains 34 parameters
- E. Don't know.

Question 24. Determine which one of the following statements is correct about Hunt's algorithm for a general classification problem using the Gini, Entropy or ClassError impurity measures and axis-aligned splits. It is evaluated using hold-out cross-validation.

- A. Hunt's algorithm always guarantees learning the decision tree with the lowest possible error on the test data.
- B. Hunt's algorithm is strictly limited to binary classification tasks.
- C. The Gini impurity is calculated using the variance of the attributes' values at each split.
- D. Hunt's algorithm determines the impurity gain based on a weighted sum of the impurity measures of the resulting branches after a split.
- E. Don't know.

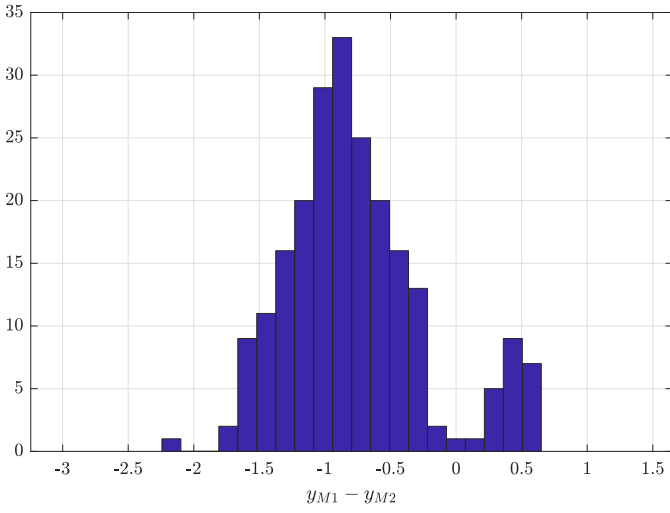


Figure 14: Histogram of $y_{M1} - y_{M2}$ based on all test observations.

Question 25. Consider a statistical procedure to determine if there is a difference in the *predictions* of two regression models, M1 and M2. We use the *paired t-test* for this purpose with the null hypothesis that there is no difference.

The paired differences in the predictions of the two models, $y_{M1} - y_{M2}$, is calculated based on the test sets in a 5-fold cross-validation setup and illustrated as a histogram in Figure 14.

Determine which combination of estimated difference ($\hat{z}$), confidence interval (CI) and p-value provide the only plausible answer.

- A. $\hat{z} = -0.71$, $CI = [-0.87, -0.71]$, $p\text{-value} < 0.05$
- B. $\hat{z} = -0.77$, $CI = [-0.84, -0.7]$, $p\text{-value} < 0.05$
- C. $\hat{z} = -0.92$, $CI = [-0.99, -0.85]$, $p\text{-value} < -0.05$
- D. $\hat{z} = 0.45$, $CI = [0.17, 0.74]$, $p\text{-value} < 0.05$
- E. Don't know.

Model 1: Logistic Regression (LR)

		Predicted	
		Class 1	Class 2
Actual	Class 1	60	73
	Class 2	148	119

Model 2: Decision Tree (DT)

		Predicted	
		Class 1	Class 2
Actual	Class 1	55	78
	Class 2	100	167

Table 6: Confusion matrices.

Question 26. Table 6 shows the confusion matrices related to a Logistic Regression model (LR) and a Decision Tree classifier (DT) trained on a particular instance of the Welding robot dataset to predict STATUS. The number of observations is $N = 400$. We define $y = 1$ to indicate a *positive outcome* and $y = 2$ to be a *negative outcome*.

Which one of the following statements is *correct*?

- A. Model 1 has lower error rate than Model 2.
- B. Model 2 has higher precision than Model 1.
- C. Predicting everything as Negative gives worse accuracy than Model 1.
- D. Model 2 has higher recall than Model 1.
- E. Don't know.

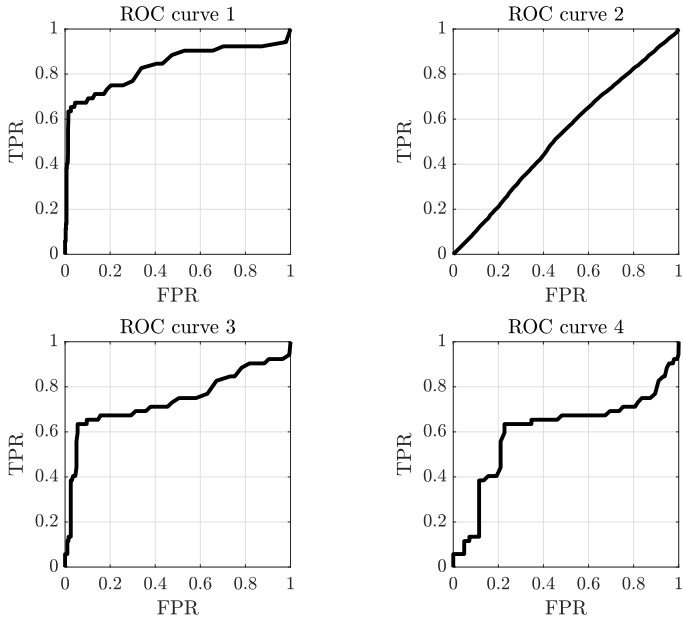


Figure 15: Candidate ROC curves for the classifier.

Question 27. We wish to predict whether an observation from the Welding robot dataset (see Table 1) belongs to the Ideal class (or not). To accomplish this, we fit a logistic regression model to the dataset, and for each observation $\mathbf{x}_i, y_i$ obtain a class-probability prediction $\hat{y}_i \in [1, 2]$. We consider $y = 1$ the *positive* outcome and $y = 2$ the *negative* outcome. We threshold the class-probability at different values θ thereby obtaining, for each value of θ , the true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN). These are plotted as functions of θ in Figure 16. Which of the receiver-operator characteristic (ROC) plots shown in Figure 15 corresponds to these graphs?

- A. ROC curve 1
- B. ROC curve 2
- C. ROC curve 3
- D. ROC curve 4
- E. Don't know.

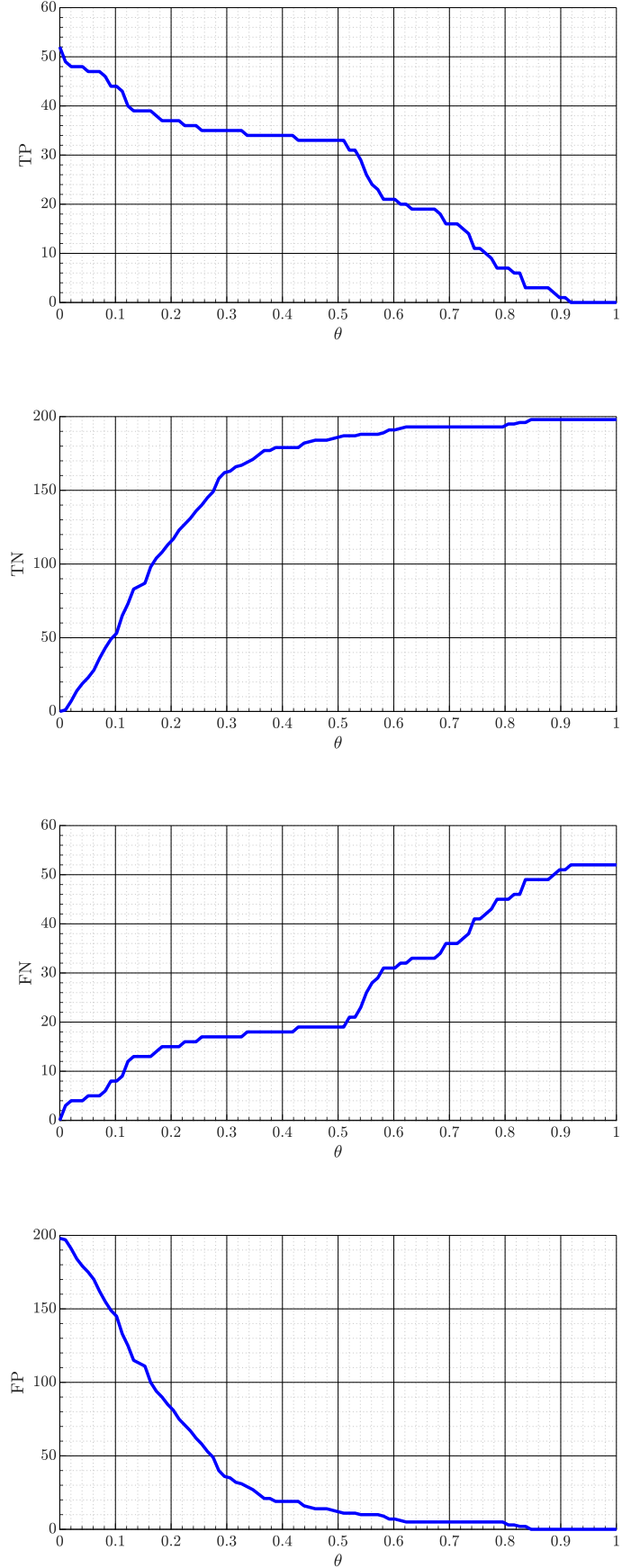


Figure 16: TP, TN, FN, and FP as functions of threshold value θ .